Project Documentation – Retail Demand & Sales Intelligence

1. Project Planning & Management

The project delivers an end-to-end retail sales intelligence system (data preparation, EDA, demand forecasting, interactive dashboard) built by five team members, with final delivery on GitHub by Tuesday 20 October 2026 (evening). Track: Artificial Intelligence & Data Science, Digital Egypt Pioneers Initiative (DEPI).

1.1 Project idea and problem statement

Retailers lose revenue when demand is estimated poorly: too little stock causes lost sales, too much ties up capital and increases waste. This project analyzes historical retail sales, forecasts future product demand with machine learning and time-series models, and presents the results in an interactive dashboard so business users can make inventory and promotion decisions.

1.2 Objectives

- Clean and prepare a retail sales dataset (missing values, outliers, features such as seasonality, holidays and promotions).
- Explore sales patterns and customer behavior through EDA and clear visualizations.
- Train and compare baseline and advanced forecasting models (e.g. Random Forest, XGBoost, Prophet) and select the best one.
- Evaluate the models with RMSE, MAE and R^2.
- Build an interactive dashboard showing sales performance, trends and forecasts.
- Deliver complete documentation, a key-findings summary and a final presentation on GitHub before the deadline.

1.3 Scope

In scope	Out of scope
Data collection and preprocessing	Real-time production API deployment
Exploratory data analysis and visualization	Live e-commerce platform integration

In scope	Out of scope
ML / time-series forecasting models	Streaming / real-time data pipelines
Model evaluation (RMSE, MAE, R^2)	
Interactive dashboard (sales KPIs, trends, forecast vs. actual)	
Documentation, findings summary, final presentation	

1.4 Deliverables

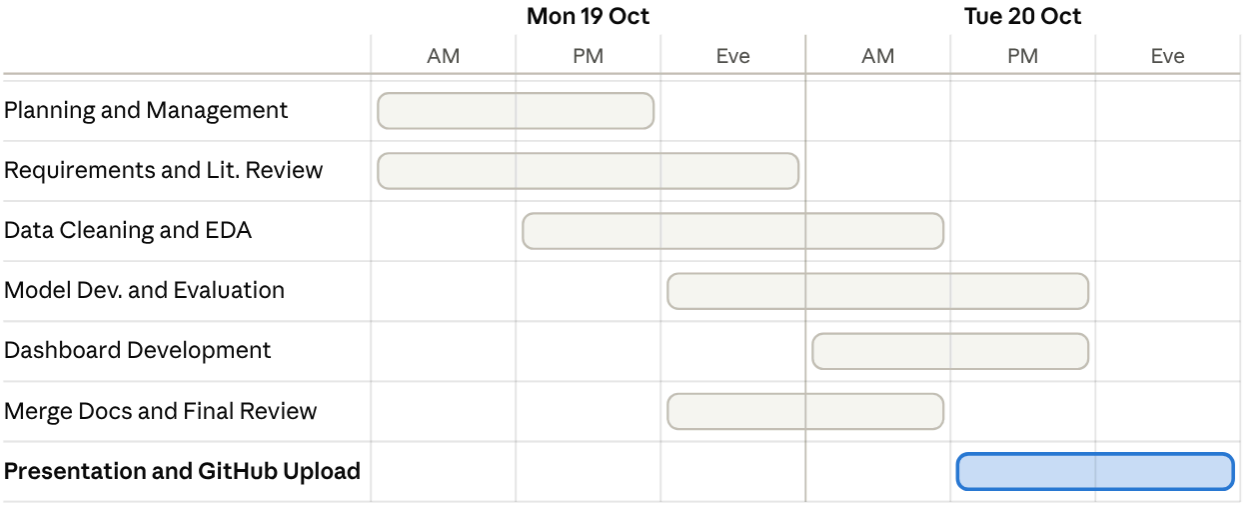
#	Deliverable	Description
1	Project documentation	Planning, literature review, requirements, analysis and design documents
2	Cleaned dataset and EDA notebook	Preprocessing pipeline and exploratory analysis with visualizations
3	Forecasting models	Trained models, comparison table and final model selection
4	Dashboard	Interactive dashboard for sales analysis and demand forecasts
5	Testing and evaluation report	Metrics, validation results and dashboard testing
6	Final presentation	Slides on problem, method, results and recommendations

1.5 Work breakdown and timeline

Phase	Activities	Output
1. Initiation & Setup	Define scope, objectives and deliverables; acquire the dataset (Rossmann Store Sales or Walmart); set up the GitHub repository	Project proposal, dataset, repo

Phase	Activities	Output
2. Requirements & Literature Review	Stakeholder analysis, user stories, functional and non-functional requirements; review of retail demand forecasting methods	Requirements and literature review documents
3. Data Preprocessing & EDA	Clean data, handle missing values and outliers, engineer temporal features, explore sales trends	Clean dataset, EDA notebook
4. Model Building & Evaluation	Build baseline and advanced models, tune, compare with RMSE / MAE / R^2, select the best	Trained model, comparison report
5. Dashboard Development	Design layout, build KPI cards, trend, product/store and forecast views, connect to model outputs	Working dashboard
6. Finalization & Delivery	Merge all documents, final testing and review, prepare presentation, upload to GitHub	Final package on GitHub

Documents merge Monday evening; Tuesday is for modeling, dashboard and delivery



Final deadline: Tue 20 Oct, evening

Gantt chart · 7 tasks, Mon AM to Tue Eve

Milestones: M1 scope, dataset and repo (Mon morning); M2 requirements and literature review (Mon afternoon); M3 cleaned data and EDA (Mon evening / Tue morning); M4 models trained and evaluated (Tue morning); M5 dashboard ready (Tue afternoon); M6 documents merged, presentation done, uploaded to GitHub (Tue evening, deadline).

1.6 Team roles and responsibilities

The team has five members; each owns one role and members cross-cover each other when a task is blocked.

#	Role	Responsibilities	Deliverable
1	Project Planning & Management	Scope, timeline, WBS, milestone tracking, risk log, KPIs, coordination, merging documents, GitHub upload	Planning document, Gantt chart, final merged package
2	Literature Review, Requirements & Final Presentation	Benchmarks, papers and forecasting techniques; stakeholders, user stories, requirements; compiling final slides	Literature review, requirements document, slide deck
3	Data Preprocessing & EDA	Cleaning, feature engineering (seasonality, holidays, promotions), exploratory analysis	Clean dataset, EDA notebook
4	Modeling & Evaluation	Baseline and advanced models, tuning, comparison with RMSE / MAE / R^2, model selection	Trained models, evaluation report
5	Dashboard, Testing & Key Findings	Dashboard design and build, connecting forecasts and KPIs, testing, key-findings summary	Interactive dashboard, testing report, key findings

Resources: Python (pandas, scikit-learn, XGBoost, Prophet, Matplotlib/Plotly), Jupyter / Google Colab, a dashboard tool (Power BI, Tableau or Streamlit), GitHub, and a public Kaggle benchmark dataset.

1.7 Risks and mitigation

Risk	Likelihood	Impact	Mitigation
Incomplete or noisy sales data	Medium	High	Use a validated benchmark dataset (Rossmann or Walmart); systematic cleaning and outlier handling
Tight deadline (final submission Tue 20 Oct)	High	High	Fast baseline model first, parallelize tasks, daily milestone check, finish documents by Tuesday morning
Poor model accuracy	Medium	Medium	Compare several models, add temporal features, tune hyperparameters, keep the best baseline as fallback
Overfitting on peak seasons	Medium	High	Time-based train/validation splits instead of random splits; regularization
Dashboard delays or tool issues	Medium	Medium	Agree on one tool early; start with KPIs + trend + forecast, improve if time allows
Team member unavailable	Medium	Medium	Cross-cover sections; keep tasks and notebooks in the shared GitHub repository
Merge conflicts or inconsistent document formatting	Medium	Low	One person merges using a shared template; fix formatting on Monday evening

1.8 Success KPIs

Category	KPI	Target
Project management	Milestones completed on time	All 6 milestones; final submission Tue 20 Oct
Project management	Deliverables uploaded to GitHub	100% of required documents

Category	KPI	Target
Data quality	Missing values after preprocessing	0% in model features
Model performance	RMSE / MAE	Lower than the naive baseline (at least 10% improvement)
Model performance	R^2 score	Higher than baseline; above 0.80 on the test set
Dashboard	Core views delivered	Sales overview, trends and seasonality, product/store analysis, forecast vs. actual
Dashboard	Usability and response time	Loads and filters in a few seconds; understandable without explanation

The numeric model targets are proposed and can be adjusted once baseline results are known.

2. Literature Review

Existing retail forecasting solutions are either cheap but descriptive (spreadsheets, BI tools) or predictive but costly and closed (enterprise suites); research and Kaggle competitions show that open-source gradient boosting with calendar and promotion features is a strong, transparent middle path, which is the approach this project takes.

2.1 Similar solutions, strengths and weaknesses

Solution	Type	Strengths	Weaknesses
Spreadsheets and traditional BI (Excel, basic Power BI / Tableau reports)	Descriptive reporting	Low cost, familiar, quick KPI views	Mostly backward-looking; manual and error-prone formulas; no real forecasting; hard to scale with many stores and products

Solution	Type	Strengths	Weaknesses
Enterprise supply-chain suites (SAP IBP, Blue Yonder, Oracle Retail)	Commercial demand planning	Enterprise-grade forecasting and inventory planning	Proprietary black-box models, long deployment, high licensing cost; out of reach for small and medium retailers
Classical statistical models (ARIMA, exponential smoothing)	Per-series forecasting	Simple, interpretable, strong baselines for single series	Struggle with many related series, promotions and external factors unless extended
Prophet	Additive time-series model	Handles trend, multiple seasonalities and holidays with little tuning	Weaker at using many store/product features and interactions than tree-based models
Gradient boosting (XGBoost, LightGBM) in Kaggle retail competitions (Rossmann, M5)	ML regression on engineered features	Top results in public retail forecasting contests; use lags, calendar, promotions and store attributes	Needs careful feature engineering and time-based validation to avoid leakage and overfitting
Deep learning (entity embeddings, DeepAR, Temporal Fusion Transformer)	Neural forecasting	Learns across many series; can model categorical features and long history	Needs more data, compute and tuning; less transparent; heavy for the project timeline

2.2 Comparison with our project

Dimension	Traditional BI / spreadsheets	Enterprise suites	Our solution
Historical and KPI analysis	Strong but manual	Comprehensive	Python EDA plus interactive dashboard (Power BI, Tableau or Streamlit)

Dimension	Traditional BI / spreadsheets	Enterprise suites	Our solution
Predictive modeling	Very limited (trendlines)	Advanced but proprietary	Open comparison of baseline, Random Forest, XGBoost and Prophet using RMSE, MAE and R^2
Cost and flexibility	Low cost, high maintenance	High investment and licenses	Open-source and modular
Transparency	Visible formulas, limited insight	Black box	Reproducible notebooks and documented models on GitHub
Real-time serving	Not available	Available	Out of scope; batch forecasts shown in the dashboard

2.3 What we take from the literature

- Start from a naive baseline and require every model to beat it, as the M4 and M5 competitions do.
- Engineer calendar, holiday, promotion and lag features; gradient boosting performs best with them in retail contests.
- Validate with time-based splits, never random splits, to avoid leakage and overfitting on peak seasons.
- Use Prophet as a complementary model for trend, seasonality and holiday effects.
- Keep deep-learning models out of scope for this timeline, but cite them as future work.

2.4 Sources

The academic references below are cited from the authors' published details and should be checked against the publisher pages before final submission.

1. Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. International Journal of Forecasting, 36(1), 54–74.
2. Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. International Journal of Forecasting, 38(4), 1346–1364.

3. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794.
4. Taylor, S. J., & Letham, B. (2018). Forecasting at scale. The American Statistician, 72(1), 37–45.
5. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
6. Bojer, C. S., & Meldgaard, J. P. (2021). Kaggle forecasting competitions: An overlooked learning opportunity. International Journal of Forecasting, 37(2), 587–603.
7. Guo, C., & Berkhahn, F. (2016). Entity Embeddings of Categorical Variables. arXiv:1604.06737.
8. Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. https://otexts.com/fpp3/
9. Kaggle. Rossmann Store Sales competition. https://www.kaggle.com/competitions/rossmann-store-sales

3. Requirements Gathering

Five stakeholder groups need to understand past sales and anticipate future demand; the system therefore must clean the data, analyze sales, forecast demand with machine learning, and present everything in an interactive dashboard, within clear quality targets.

3.1 Stakeholders and their needs

Stakeholder	Need
Retail store managers	Monitor sales performance; identify best-selling and slow-moving products
Business owners	Make strategic decisions that improve profitability and overall performance
Inventory managers	Plan stock levels; reduce stockouts and excess inventory
Sales and marketing teams	Understand sales trends; find products that may benefit from promotions
Data analysts	Analyze data and produce insights that support decisions

3.2 User needs and user stories

- As a store manager, I want to see best-selling products and highest-revenue products, so I can focus on what sells.
- As an inventory manager, I want forecasts of future demand and seasonal periods, so I can plan replenishment and avoid stockouts.
- As a marketing team member, I want to see sales trends and slow movers, so I can choose products for promotions.
- As a business owner, I want KPIs and forecasts in one interactive dashboard, so I can decide without technical help.
- As a data analyst, I want a clean dataset and reproducible notebooks, so I can verify and extend the analysis.

3.3 Functional requirements

ID	Requirement	Description
FR-1	Data collection	Collect sales, product, pricing and inventory data from available sources
FR-2	Data preprocessing	Clean data; handle missing values, duplicates and outliers
FR-3	Exploratory data analysis	Explore patterns, trends and relationships among variables
FR-4	Sales performance analysis	Analyze total sales, revenue, best-selling products and store/branch performance when branch data exists
FR-5	Demand forecasting	Build machine learning models that forecast future product demand from historical sales
FR-6	Inventory insights	Give data-driven insights on stock levels, stockouts and overstocking, where data is available
FR-7	Interactive dashboard	Power BI, Tableau or Streamlit dashboard with KPIs, sales trends and forecasting results
FR-8	Model evaluation	Evaluate with MAE, RMSE (and R^2); compare models and select the best

3.4 Non-functional (quality) requirements

ID	Quality	Requirement	Target
NFR-1	Accuracy	Appropriate accuracy in analysis and forecasting, checked with statistical metrics	Beat the naive baseline by at least 10% on RMSE / MAE; R^2 above 0.80
NFR-2	Performance	Process data and show reports in reasonable time for the dataset size	Dashboard loads and filters in a few seconds
NFR-3	Usability	Clear dashboard understandable by non-technical users	Usable without explanation
NFR-4	Security	Protect sales and customer data if present; restrict sensitive data by user permission	No sensitive data in public repo; access limited to permitted users
NFR-5	Scalability	Able to grow to larger datasets and more products or branches	Pipeline not hard-coded to one store or product
NFR-6	Reliability	Consistent, reproducible results; errors and missing data handled	Fixed random seeds; 0% missing values in model features
NFR-7	Maintainability	Organized, documented code so data, models and features can be updated	Notebooks documented and kept in GitHub

3.5 Data requirements

Field	Description	Priority
Product ID	Unique product identifier	Required
Date	Date of the sales transaction	Required
Sales quantity	Number of units sold	Required
Unit price	Price per unit	Required
Revenue	Total revenue, or computable from quantity and price	Required
Product category	Category of the product	Useful

Field	Description	Priority
Store / Branch ID	Store identifier	If available
Inventory level	Current or historical stock level	If available
Promotional information	Discounts or promotions	If available

Suggested dataset: Rossmann Store Sales (Kaggle, about 1M rows x 18 columns after merge, target Sales, regression); Walmart is the alternative named in the plan.

3.6 Tools and technologies

Tool	Use
Python	Data analysis and model development
Pandas, NumPy	Data manipulation and cleaning
Matplotlib, Seaborn	Charts and EDA
Scikit-learn, XGBoost, Prophet	Machine learning, forecasting and evaluation
Power BI / Tableau / Streamlit	Interactive dashboards
Jupyter Notebook / Google Colab	Experimentation and analysis
GitHub	Version control and final submission

3.7 Expected outcomes

- A comprehensive analysis of sales performance and trends.
- Identification of products and periods with the highest sales.
- Forecasts of future demand to support inventory planning.
- Actionable insights for data-driven decisions.
- Reduced risk of stockouts or excess inventory when inventory data is available.
- An interactive dashboard with KPIs, analysis results and forecasts.